AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH
Department of Computer Science

Project Report

RepRush

Android Gym Management & Fitness Tracking Application

Course: Mobile Application Development [A]
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Group Information

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1. Project Overview

RepRush is an Android application that serves as a complete gym management and fitness tracking system. It addresses two problems in a single platform: gym owners who manage member records, payments, and attendance manually, and gym members who lack structured workout programming and progress tracking.

Key Features at a Glance

- **AI-generated workout plans:** using Google Gemini 1.5 Flash API
- **Full exercise library:** sourced from free-exercise-db, cached locally for offline use
- **Workout session logging:** with rest timer, auto-save, and crash recovery
- **Progress analytics:** including a GitHub-style volume heatmap, strength score, and body weight tracking
- **Gamification:** points, streaks, 11 achievement badges, and a monthly leaderboard
- **Admin dashboard:** for member management, billing, attendance, and revenue reporting

Tech Stack

Category	Technology
Language & Architecture	Kotlin, MVVM with LiveData
Local Database	Room (SQLite) — 16 entities
Cloud Database	Firebase Firestore with offline persistence
Authentication	Firebase Auth — Google Sign-In
Dependency Injection	Hilt
AI Integration	Google Gemini 1.5 Flash API
Exercise Data	free-exercise-db (open-source, bundled JSON + images)
Charts & Heatmap	MPAndroidChart, Kizitonwose HeatMapCalendar
Image Loading	Glide
UI Framework	Material 3 (DayNight theme)
Min / Target SDK	API 26 (Android 8.0) / API 37 (Android 15)

2. Functional Requirements

Type of Users

- **Admin:** Pre-created in Firestore. Manages all business operations. No access to member fitness features.
- **Member:** Registers via Google Sign-In. Starts in a pending state until approved by an admin with a membership package. Suspended or expired members are locked out until reactivated.

Common Functionality

- **Google Sign-In:** via Firebase Authentication with automatic role detection and routing
- **In-app notification inbox:** for membership updates, payment receipts, and announcements
- **Offline access:** all member fitness features work fully without an internet connection
- **Light and dark mode:** via Material 3 DayNight theme

User Specific Functionality

Admin

Member Management:

- View pending registrations and approve with a package or reject
- Register walk-in members manually
- Browse member directory with search and status filters (Active, Pending, Expired, Suspended, Rejected)
- View any member's profile — payment history, attendance log, and workout stats
- Suspend, reactivate, or permanently remove accounts

Package Management:

- Create, edit, and deactivate membership packages
- Assign packages to members during approval or renewal

Payments & Receipts:

- Record payments with amount, method (cash, bank transfer, mobile banking), and date
- System auto-generates a digital receipt delivered to the member's notification inbox
- View full payment log filterable by date range and member
- Void incorrect payments with a reason note

Attendance & Dashboard:

- Mark daily attendance for any member

- Dashboard shows six live metrics: active members, pending requests, expiring soon, checked in today, monthly revenue, yearly revenue
- Post gym-wide announcements visible to all active members
- Configure grace period (in days) before auto-suspension of expired members

Member

Exercise Library:

- Full exercise catalog downloaded on first launch and cached locally
- Searchable by name, filterable by muscle group and equipment
- Exercise detail page shows image, personal record history, estimated 1RM, and progress chart
- Custom exercises can be added manually

AI Workout Plan Generation:

- Eight-step questionnaire: goal, training days, split type, session duration, equipment, fitness level, plan duration, injuries
- Answers and matching exercise names sent to Gemini 1.5 Flash API
- JSON response validated and imported into Room
- Multiple plans can be saved; one is designated as active

Workout Logging:

- Start from the active plan (exercises pre-loaded) or as a blank session
- Log each set with weight and reps; mark sets as warmup to exclude from points and records
- Rest timer appears after each completed set and runs in a foreground notification while screen is off
- Session auto-saves every 30 seconds; interrupted sessions can be resumed on next app open

Progress & Gamification:

- Personal records detected automatically per exercise per rep count on session completion
- Points per session: 50 for attendance, 10 per exercise, 2 per set, 100 per new record (500 daily cap)
- Streak bonuses: +200 pts at 7 days, +1000 pts at 30 days
- 11 achievement badges with specific unlock conditions
- Monthly leaderboard ranking opted-in members by points
- GitHub-style heatmap showing 6 months of workout volume — cell intensity based on total lifting volume
- Progress tab with three sub-tabs:
  - Overview — heatmap and full workout history
  - Body — daily weight log with rolling-average chart
  - Strength — Strength Score (sum of 1RM across 5 compound lifts), per-lift trends, 12-week volume chart

3. Challenges & Difficulties

- **Room and KSP version incompatibility:** Room 2.6.1 failed to compile with KSP 2.3.7 due to an IllegalStateException when processing suspend DAO functions. Resolved by upgrading Room to 2.7.1.

- **Gemini API key misconfiguration:** The key was placed in local.properties but the build script read from gradle.properties, causing BuildConfig.GEMINI_API_KEY to be empty at runtime. Errors were silently swallowed with no feedback. Fixed by moving the key to the correct file and adding explicit error logging.

- **Firestore composite indexes missing:** Queries combining a field filter with orderBy on a different field threw FAILED_PRECONDITION errors at runtime. Indexes were created in the Firebase Console and a firestore.indexes.json file was established covering all required indexes across all milestones.

- **Firestore field name mismatch:** The MembershipPackage data class used isActive, but Firestore stored the field as active. The query whereEqualTo("isActive", true) silently returned zero results. Renamed the field to active throughout the codebase.

- **LiveData stickiness:** ViewModels scoped to the activity retained stale operationResult values. Newly opened Fragments immediately received the previous result, causing unintended navigation and duplicate Snackbars. Fixed by making operationResult a nullable LiveData initialised to null, adding clearOperationResult(), and null-checking in all observers.

- **Android 15 edge-to-edge rendering:** With targetSdk 37, the OS draws the app behind the status bar and notch by default. All host Activity files lacked inset handling, so content was hidden behind system UI on every screen. Fixed by applying a ViewCompat.setOnApplyWindowInsetsListener at the activity level consuming both systemBars and displayCutout insets.

- **Hardcoded JDK path:** A machine-specific org.gradle.java.home path in gradle.properties caused Gradle to fail on the second team member's machine. Removed the line so Android Studio uses its bundled JDK automatically on both machines.

- **Wger API replaced by free-exercise-db:** The original design relied on the Wger REST API. External dependency and reliability concerns led to replacing it with the free-exercise-db open-source dataset. Exercise JSON is now bundled in app assets and images are downloaded on first launch to local storage. The muscleImageUrl field is always null as the dataset includes no muscle diagrams.

- **Rest timer bottom sheet not appearing:** During Milestone 5 testing, the rest timer BottomSheetDialogFragment did not appear after set completion despite the foreground service being correctly implemented. The issue is a timing and state management problem between the set completion event and the sheet display trigger. Currently under investigation.

- **Bodyweight toggle layout break:** The bodyweight toggle in session set rows incorrectly keeps a label visible while shifting the reps input field, breaking the row layout. Pending resolution in Milestone 5.

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